



**Internship Report**  
**On**  
**CAR PRICE PREDICTION USING MACHINE LEARNING**

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## 2. Student's Declaration

I, Ankita Pokhriyal ,a student of B.Tech program, Roll No. 240970101021 of the Department of Computer Science & Engineering, THDC- Institute of Hydropower Engineering & Technology College do hereby declare that I have completed the mandatory internship in EISystems Technologies.

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## 6. Executive Summary

The internship provided practical exposure to Data Science, Machine Learning, Python programming, data analysis, model building, and model deployment. The main objective of the internship was to develop a basic understanding of the complete machine learning process, from preparing data to training, evaluating, and deploying a model. Along with technical skills, the internship also helped in understanding project development, documentation, and submission procedures.

During the training phase, I learned the fundamentals of Data Science, Artificial Intelligence, and Machine Learning, including different types of machine learning and the machine learning lifecycle. I strengthened my Python skills by working with lists, tuples, dictionaries, sets, loops, and functions. I also learned NumPy for working with arrays and Pandas for handling and analysing datasets. Various Pandas operations such as filtering, grouping, sampling, merging, handling missing values, and categorical data encoding were practiced using datasets such as Netflix and Titanic.

The internship also covered data visualization using Matplotlib, where I created line, scatter, bar, pie, and subplot graphs. I learned the basics of linear regression and classification techniques and practiced machine learning using datasets and image data. Practical sessions included handwritten digit recognition using Logistic Regression, Iris classification using KNN, and image classification using SVM. Concepts such as train-test splitting, model evaluation, cross-validation, and parameter tuning were also introduced.

During the project phase, I applied the concepts learned during training to develop and test a machine learning project. The project involved preparing the dataset, performing the required data preprocessing, training the model, testing its performance, and obtaining predictions. I also learned how to save a trained model as a pickle file and deploy it using Streamlit so that the model could be used through an application interface.

Towards the end of the internship, the instructor explained the project report preparation and submission process, provided guidance regarding the required format and instructions, and cleared the doubts raised by the interns. Overall, the internship helped me gain practical knowledge of the machine learning workflow, improve my Python and data-handling skills, and understand how a machine learning project is developed from data preparation to deployment.

## 7. Overview of Organization

EISystems Services (EISys) is a leading Indian technology organization with operations across India. It provides technical training and workshops for enterprises and the student community. The organization offers training in various fields such as Cybersecurity, Machine Learning, Automobiles, Internet of Things (IoT), Robotics, and Social Media. Since its founding, EISystems has trained approximately 50,000 students and impacted around 2 lakh students through its outreach initiatives. The information provided on the official About page does not give separate formal statements of the organization's Vision, Mission, and Values. The provided About page does not specify a separate formal internship policy. During the internship, the intern's role involved technical training, practical learning, implementation of Machine Learning concepts, and project development. The internship provided an opportunity to apply theoretical knowledge to a practical project using tools and technologies related to Data Science and Machine Learning.

Some of the colleges where the EISystems had already felt our presence are given below:-

Indian Institute of Science, Bangalore  
Indian Institute of Technology, Bombay  
Indian Institute of Technology, Delhi  
Indian Institute of Technology, Madras  
Indian Institute of Technology, Kanpur  
Indian Institute of Technology, Roorkee  
Indian Institute of Technology, Guwahati  
Indian Institute of Technology (Banaras Hindu University), Varanasi  
Indian Institute of Technology, Indore  
Indian Institute of Technology, Jodhpur  
Indian Institute of Technology, Hyderabad  
National Institute of Technology, Tiruchirappalli  
National Institute of Technology, Warangal  
National Institute of Technology, Calicut  
National Institute of Technology, Patna  
National Institute of Technology, Jalandhar  
National Institute of Technology, Jaipur  
National Institute of Technology, Durgapur  
National Institute of Technology, Surat  
National Institute of Technology, Allahabad  
Indian Institute of Information Technology, Allahabad  
ABV Indian Institute of Information Technology, Gwalior  
PDP Indian Institute of Information Technology, Jabalpur  
Jawahar Lal Nehru Technological Univeristy, Hyderabad  
College of Engineering, Guindy  
Delhi Technological Univeristy, New Delhi  
& around 100 engineering colleges.

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## 8. Project Summary

### 8.1 Idea Behind Making This Project

The idea behind this project is to develop a Machine Learning-based system that can estimate the selling price of a used car based on its different characteristics. The price of a used car depends on several factors such as its brand, manufacturing year, kilometres driven, fuel type, seller type, transmission, ownership, mileage, engine capacity, maximum power, and number of seats.

Manual estimation of a car's price can be difficult because many factors affect its value. Therefore, this project uses historical car data to train a Machine Learning model that can learn the relationship between these features and the selling price.

The project was developed as a practical application of the concepts learned during the AI/ML internship.

### 8.2 About the Project

The project titled “Car Price Prediction Using Machine Learning” is a Machine Learning application developed using Python. The dataset used in the project was obtained from Kaggle and contains information about used cars.

The project uses the following input features for prediction:

- Car brand/name
- Manufacturing year
- Kilometres driven
- Fuel type
- Seller type
- Transmission
- Owner
- Mileage
- Engine capacity
- Maximum power
- Number of seats

The target variable of the project is selling\_price.

The categorical features such as car brand, fuel type, seller type, transmission, and owner are converted into numerical values during preprocessing. Numerical information such as mileage, engine and maximum power is cleaned and converted into numerical form.

After preprocessing, the dataset is divided into training and testing data. A Linear Regression Machine Learning model is then trained using the training data. The trained model is used to predict car prices and is finally saved as model.pkl.

The saved model is integrated into a Streamlit application. The application provides an interactive interface where a user can select the required car details and click the Predict button to obtain an estimated car price.

### 8.3 Software Used in the Project

The following software, programming language, libraries and tools were used:

| Software / Tool  | Purpose                                            |
|------------------|----------------------------------------------------|
| Python           | Programming and Machine Learning                   |
| Jupyter Notebook | Data preprocessing, analysis and model development |
| Pandas           | Loading and manipulating the dataset               |
| NumPy            | Numerical operations                               |
| Scikit-learn     | Train-test splitting and Linear Regression         |
| Pickle           | Saving the trained Machine Learning model          |
| Streamlit        | Creating the interactive web application           |
| Kaggle           | Source of the car dataset                          |

The Streamlit program imports Pandas, NumPy, Pickle and Streamlit, loads the saved model.pkl, and reads the Car details.csv dataset.

## 8.4 Technical Apparatus Requirements

### Hardware Requirements

The project can be developed and executed using a standard computer or laptop with:

- Processor: Any modern processor
- RAM: Minimum 4 GB recommended
- Storage: Sufficient space for Python, dataset and project files
- Keyboard and mouse
- Internet connection for downloading the dataset and required software/packages

### Software Requirements

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- Pickle
- Web browser
- Kaggle dataset

The project requires the dataset file Car details.csv and the trained model file model.pkl for the Streamlit application to work.

## 8.5 Result / Working of the Project

The project successfully trains a Linear Regression model using the processed car dataset. In the Jupyter Notebook, the dataset is first loaded and preprocessed. The torque column is removed, missing values are dropped, duplicate records are removed, and required columns are converted into suitable numerical forms.

The car brand is extracted from the car name. For example, the first part of the car name is used as the brand. The categorical features are then encoded into numerical values. After preprocessing, selling\_price is separated as the

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output variable, while the remaining features are used as input variables.

The data is divided into training and testing sets using `train_test_split()`. A Linear Regression model is created and trained using:

```
model.fit(train_x, train_y)
```

The model then generates predictions using:

```
model.predict(test_x)
```

A sample input is also provided to the trained model to verify that it can generate a car price prediction.

Finally, the trained model is saved as `model.pkl` using Python's Pickle module.

In the Streamlit application, the saved model is loaded and the user provides the required car details. These inputs are converted into the numerical representation used during model training. When the user clicks the Predict button, the processed input is passed to the trained model and the estimated car price is displayed on the screen.

## 8.6 Research Done and Process Followed

The project involved practical research and learning related to Machine Learning-based price prediction. The process involved understanding the structure of the Kaggle car dataset, identifying the relevant input features, checking missing values and duplicate records, cleaning numerical columns, and converting categorical features into numerical values.

The Machine Learning workflow was then studied and implemented using a training and testing dataset. Linear Regression was selected and implemented using Scikit-learn. The trained model was tested by generating predictions on the test data and by providing a sample input.

The final step involved learning how to save the trained Machine Learning model using Pickle and integrate it into a Streamlit application for interactive prediction.

## 8.7 Overall Outcome

The completed project provides a working interface for estimating the selling price of a used car. It demonstrates the complete basic Machine Learning workflow, beginning with dataset preparation and preprocessing, followed by model training and prediction, and finally integration with a Streamlit-based user interface.

## 9. Data Flow Diagram / Process Flow

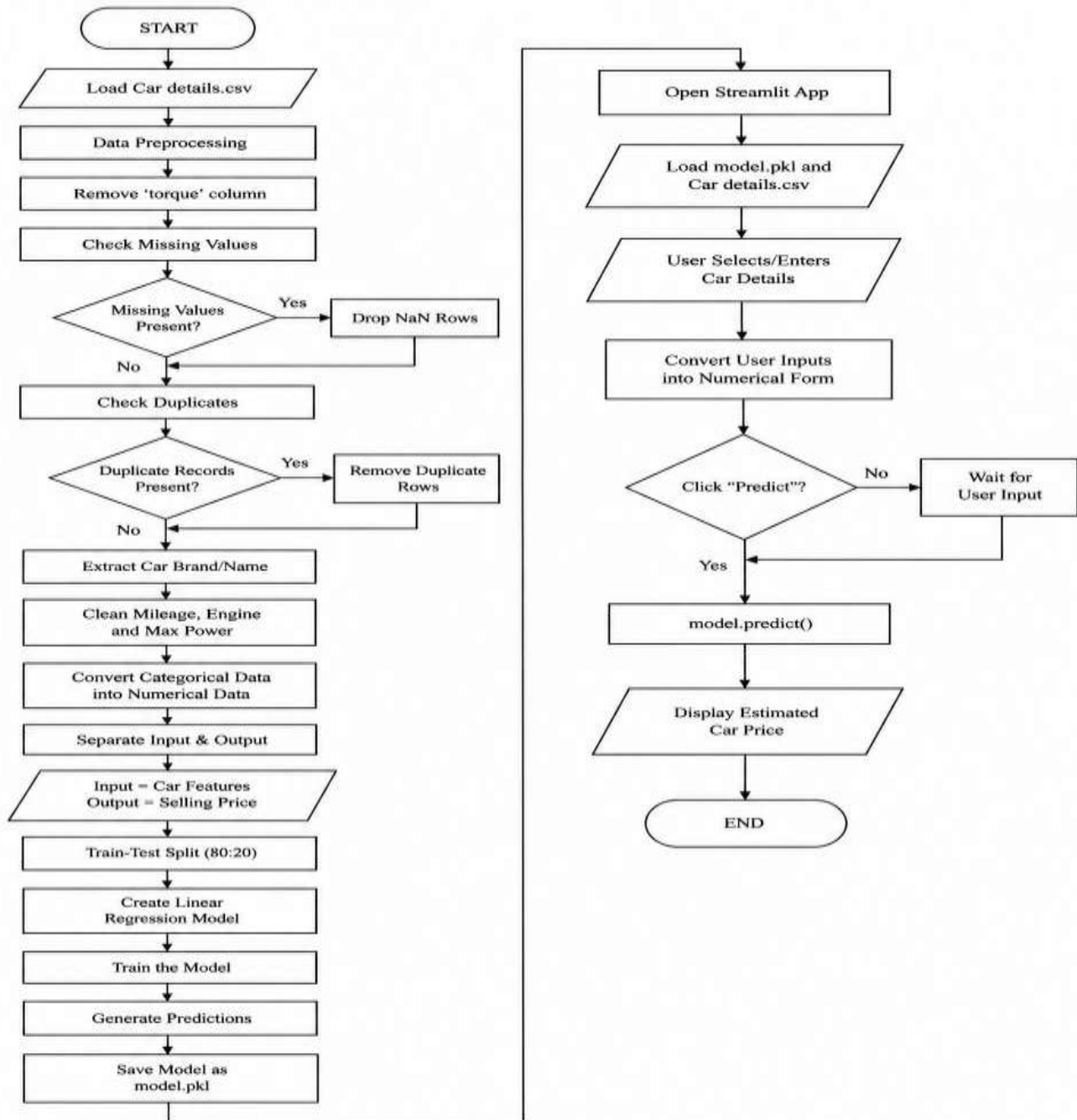


Fig.9.1 Overall Process Flow of the Car Price Prediction System

## 10. Code / Program with Supported Screenshots

```
[1]: import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

[2]: cars_data = pd.read_csv("Car_details.csv")

[3]: cars_data.head()
```

|   | name                         | year | selling_price | km_driven | fuel   | seller_type | transmission | owner        | mileage    | engine  | max_power  | torque                   | seats |
|---|------------------------------|------|---------------|-----------|--------|-------------|--------------|--------------|------------|---------|------------|--------------------------|-------|
| 0 | Maruti Swift Dzire VDI       | 2014 | 450000        | 145500    | Diesel | Individual  | Manual       | First Owner  | 23.4 kmpl  | 1248 CC | 74 bhp     | 190Nm@ 2000rpm           | 5.0   |
| 1 | Skoda Rapid 1.5 TDI Ambition | 2014 | 370000        | 120000    | Diesel | Individual  | Manual       | Second Owner | 21.14 kmpl | 1498 CC | 103.52 bhp | 250Nm@ 1500-2500rpm      | 5.0   |
| 2 | Honda City 2017-2020 EXI     | 2006 | 158000        | 140000    | Petrol | Individual  | Manual       | Third Owner  | 17.7 kmpl  | 1497 CC | 78 bhp     | 12.7@ 2700(kgm@ rpm)     | 5.0   |
| 3 | Hyundai i20 Sportz Diesel    | 2010 | 225000        | 127000    | Diesel | Individual  | Manual       | First Owner  | 23.0 kmpl  | 1396 CC | 90 bhp     | 22.4 kgm at 1750-2750rpm | 5.0   |
| 4 | Maruti Swift VXI BSIII       | 2007 | 130000        | 120000    | Petrol | Individual  | Manual       | First Owner  | 16.1 kmpl  | 1298 CC | 88.2 bhp   | 11.5@ 4500(kgm@ rpm)     | 5.0   |

```
[4]: cars_data.drop(columns=['torque'], inplace=True)

[5]: cars_data.head()
```

|   | name                   | year | selling_price | km_driven | fuel   | seller_type | transmission | owner       | mileage   | engine  | max_power | seats |
|---|------------------------|------|---------------|-----------|--------|-------------|--------------|-------------|-----------|---------|-----------|-------|
| 0 | Maruti Swift Dzire VDI | 2014 | 450000        | 145500    | Diesel | Individual  | Manual       | First Owner | 23.4 kmpl | 1248 CC | 74 bhp    | 5.0   |

Fig 10.1 Importing libraries & Loading dataset

```
[5]:
```

|   | name                         | year | selling_price | km_driven | fuel   | seller_type | transmission | owner        | mileage    | engine  | max_power  | seats |
|---|------------------------------|------|---------------|-----------|--------|-------------|--------------|--------------|------------|---------|------------|-------|
| 0 | Maruti Swift Dzire VDI       | 2014 | 450000        | 145500    | Diesel | Individual  | Manual       | First Owner  | 23.4 kmpl  | 1248 CC | 74 bhp     | 5.0   |
| 1 | Skoda Rapid 1.5 TDI Ambition | 2014 | 370000        | 120000    | Diesel | Individual  | Manual       | Second Owner | 21.14 kmpl | 1498 CC | 103.52 bhp | 5.0   |
| 2 | Honda City 2017-2020 EXI     | 2006 | 158000        | 140000    | Petrol | Individual  | Manual       | Third Owner  | 17.7 kmpl  | 1497 CC | 78 bhp     | 5.0   |
| 3 | Hyundai i20 Sportz Diesel    | 2010 | 225000        | 127000    | Diesel | Individual  | Manual       | First Owner  | 23.0 kmpl  | 1396 CC | 90 bhp     | 5.0   |
| 4 | Maruti Swift VXI BSIII       | 2007 | 130000        | 120000    | Petrol | Individual  | Manual       | First Owner  | 16.1 kmpl  | 1298 CC | 88.2 bhp   | 5.0   |

```
[6]: cars_data.shape

(5): (5128, 12)

Preprocessing

NULL CheckK

[7]: cars_data.isnull().sum()
```

```
[7]: name          0
year          0
selling_price  0
km_driven     0
fuel          0
seller_type   0
transmission  0
owner         0
mileage       221
engine        221
max_power     215
seats         221
dtype: int64
```

Fig 10.2 Removing torque

```
[8]: cars_data.dropna(inplace=True)

[9]: cars_data.shape

[9]: (7907, 12)

Duplicate Check

[10]: cars_data.duplicated().sum()

[10]: 91164(1189)

[11]: cars_data.drop_duplicates(inplace=True)

[12]: cars_data.shape

[12]: (6718, 12)

[13]: cars_data

[13]:
```

|   | name                         | year | selling_price | km_driven | fuel   | seller_type | transmission | owner        | mileage    | engine  | max_power  | seats |
|---|------------------------------|------|---------------|-----------|--------|-------------|--------------|--------------|------------|---------|------------|-------|
| 0 | Maruti Swift Dzire VDI       | 2014 | 450000        | 145500    | Diesel | Individual  | Manual       | First Owner  | 23.4 kmpl  | 1248 CC | 74 bhp     | 5.0   |
| 1 | Skoda Rapid 1.5 TDI Ambition | 2014 | 370000        | 120000    | Diesel | Individual  | Manual       | Second Owner | 21.14 kmpl | 1498 CC | 103.52 bhp | 5.0   |
| 2 | Honda City 2017-2020 EXi     | 2006 | 158000        | 140000    | Petrol | Individual  | Manual       | Third Owner  | 17.7 kmpl  | 1497 CC | 78 bhp     | 5.0   |
| 3 | Hyundai i20 Sportz Diesel    | 2010 | 225000        | 127000    | Diesel | Individual  | Manual       | First Owner  | 23.0 kmpl  | 1396 CC | 90 bhp     | 5.0   |

Fig 10.3 Duplicate checking and drop\_duplicates()

```
8 mileage      6718 non-null  str
9 engine       6718 non-null  str
10 max_power   6718 non-null  str
11 seats       6718 non-null  float64
dtypes: float64(1), int64(3), str(8)
memory usage: 1.2 MB

Data Analysis

[25]: for col in cars_data.columns:
      print('unique values of ' + col)
      print(cars_data[col].unique())
      print("=====")

Unique values of name
<ArrayStringArray>
[
    'Maruti Swift Dzire VDI',
    'Skoda Rapid 1.5 TDI Ambition',
    'Honda City 2017-2020 EXi',
    'Hyundai i20 Sportz Diesel',
    'Maruti Swift VXI BSIII',
    'Hyundai Xcent 1.2 VVI & Plus',
    'Maruti Wagon R LXI DDD BSIII',
    'Maruti 800 GX BSII',
    'Toyota Etios VXD',
    'Ford Figo Diesel Celebration Edition',
    ...,
    'Toyota Innova 2.5 E 7 STR',
    'Honda City ZXI AT',
    'Ford Fiesta 1.4 Duratorq EXi',
    'Chevrolet Cruze LT',
    'Maruti Celerio ZXI AMT BSIV',
    ...
]

[26]: def get_brand_name(car_name):
      car_name = car_name.split(' ')[0]
      return car_name.strip()
```

Fig 10.4 Brand extraction

```
[17]: def clean_data(value):
      value = value.split(' ')[0]
      value = value.strip()
      if value == '':
          value = 0
      return float(value)

[18]: get_brand_name('Maruti Swift DZire VDI')

[19]: 'Maruti'

[19]: cars_data['name'] = cars_data['name'].apply(get_brand_name)

[20]: cars_data['name'].unique()

[20]: <ArrowStringArray>
      [
        'Maruti',      'Skoda',      'Honda',      'Hyundai',
        'Toyota',      'Ford',      'Renault',    'Mahindra',
        'Tata',        'Chevrolet', 'Datsun',     'Jeep',
        'Mercedes-Benz', 'Mitsubishi', 'Audi',       'Volkswagen',
        'BMW',         'Nissan',    'Lexus',     'Jaguar',
        'Land',        'MG',       'Volvo',     'Daewoo',
        'Kia',         'Fiat',     'Force',     'Ambassador',
        'Ashok',      'Isuzu',    'Opel'
      ]
      Length: 31, dtype: str

[21]: cars_data['mileage'] = cars_data['mileage'].apply(clean_data)

[22]: cars_data['max_power'] = cars_data['max_power'].apply(clean_data)

[23]: cars_data['engine'] = cars_data['engine'].apply(clean_data)

[24]: for col in cars_data.columns:
      print('Unique values of ' + col)
```

Fig 10.5 Cleaning mileage, engine and max power

```
=====
Unique values of seats
[ 5.  4.  7.  8.  6.  9. 10. 14.  2.]
=====

]: import warnings
   warnings.filterwarnings("ignore")

]: cars_data['name'] = cars_data['name'].replace({
      'Maruti': 1, 'Skoda': 2, 'Honda': 3, 'Hyundai': 4, 'Toyota': 5,
      'Ford': 6, 'Renault': 7, 'Mahindra': 8, 'Tata': 9,
      'Chevrolet': 10, 'Datsun': 11, 'Jeep': 12, 'Mercedes-Benz': 13,
      'Mitsubishi': 14, 'Audi': 15, 'Volkswagen': 16, 'BMW': 17,
      'Nissan': 18, 'Lexus': 19, 'Jaguar': 20, 'Land': 21, 'MG': 22,
      'Volvo': 23, 'Daewoo': 24, 'Kia': 25, 'Fiat': 26, 'Force': 27,
      'Ambassador': 28, 'Ashok': 29, 'Isuzu': 30, 'Opel': 31
    })

]: cars_data['transmission'].unique()

]: <ArrowStringArray>
   ['Manual', 'Automatic']
   Length: 2, dtype: str

]: cars_data['transmission'] = cars_data['transmission'].replace({
      'Manual': 1,
      'Automatic': 2
    })

]: cars_data['seller_type'].unique()

]: <ArrowStringArray>
   ['Individual', 'Dealer', 'Trustmark Dealer']
   Length: 3, dtype: str

]: cars_data['seller_type'] = cars_data['seller_type'].replace({
```

Fig 10.6 a. Encoding brand/fuel/seller/transmission/owner

```
[30]: cars_data['seller_type'] = cars_data['seller_type'].replace({
      'Individual': 1,
      'Dealer': 2,
      'Trustmark Dealer': 3
    })

[31]: cars_data.info()

<class 'pandas.DataFrame'>
Index: 6718 entries, 0 to 8125
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   name             6718 non-null   object  
1   year             6718 non-null   int64   
2   selling_price    6718 non-null   int64   
3   km_driven        6718 non-null   int64   
4   fuel             6718 non-null   str      
5   seller_type      6718 non-null   object  
6   transmission     6718 non-null   object  
7   owner            6718 non-null   str      
8   mileage          6718 non-null   float64  
9   engine           6718 non-null   float64  
10  max_power        6718 non-null   float64  
11  seats            6718 non-null   float64  
dtypes: float64(4), int64(3), object(3), str(2)
memory usage: 798.4+ KB

[32]: cars_data['fuel'].unique()

[32]: <ArrowStringArray>
      ['Diesel', 'Petrol', 'LPG', 'CNG']
      Length: 4, dtype: str

[33]: cars_data['fuel'] = cars_data['fuel'].replace({
      'Diesel': 1,
      'Petrol': 2,
```

Fig 10.6 b. Encoding brand/fuel/seller/transmission/owner

```
-----
seats
[ 5, 4, 7, 8, 6, 9, 10, 14, 2.]

[41]: cars_data.isnull().sum()

[41]: name             0
      year            0
      selling_price    0
      km_driven        0
      fuel             0
      seller_type      0
      transmission     0
      owner            0
      mileage          0
      engine           0
      max_power        0
      seats            0
      dtype: int64

[42]: input_data = cars_data.drop(columns=['selling_price'])
      output_data = cars_data['selling_price']

[43]: train_x, test_x, train_y, test_y = train_test_split(input_data, output_data, test_size=0.2)

      Model Creation

[44]: model = LinearRegression()

      Train Model

[45]: model.fit(train_x, train_y)

[46]: LinearRegression
```

Fig 10.7 Separating input and output & Train-test split & Model training

```
[45]: > linearRegression <- >
      Parameters
      Fitted attributes

[46]: predict = model.predict(test_x)

[47]: predict
array([[ 353447.82619126,  113827.45233336,  361287.72181465, ...,
        455297.47514653,  671842.27962556, -187285.27854831],
      shape=(1344,))

[48]: train_x.head(1)

[49]:
   name  year  km_driven  fuel  seller_type  transmission  owner  mileage  engine  max_power  seats
5807    4   2015    35000    2           1             1       2      16.5   1197.0      81.83    5.0

[50]: input_data_model = pd.DataFrame(
      [[5,2022,12000,1,1,1,12.99,2494.0,100.6,5.0]],
      columns=['name','year','km_driven','fuel','seller_type','transmission','owner','mileage','engine','max_power','seats'])

[51]: input_data_model

[52]:
   name  year  km_driven  fuel  seller_type  transmission  owner  mileage  engine  max_power  seats
0      5   2022    12000    1           1             1       1      12.99   2494.0      100.6    5.0
```

Fig 10.8 Linear Regression model

```
columns=['name','year','km_driven','fuel','seller_type','transmission','owner','mileage','engine','max_power','seats'])

[50]: input_data_model

[51]:
   name  year  km_driven  fuel  seller_type  transmission  owner  mileage  engine  max_power  seats
0      5   2022    12000    1           1             1       1      12.99   2494.0      100.6    5.0

[52]: model.predict(input_data_model)

[53]: array([1051867.25647825])

[54]: import pickle as pk

[55]: pk.dump(model,open('model.pkl','wb'))

[ ]:

[ ]:

[ ]:
```

Fig 10.9 Prediction & Saving model.pkl

```

1 import pandas as pd
2 import numpy as np
3 import pickle as pk
4 import streamlit as st
5 # Load Trained model
6 model = pk.load(open('model.pkl', 'rb'))
7 st.header('Car Price Prediction ML Model')
8 # Load dataset
9 cars_data = pd.read_csv('Car details.csv')
10 # Function to get car brand
11 def get_brand_name(car_name):
12     car_name = car_name.split(' ')[0]
13     return car_name.strip()
14 cars_data['name'] = cars_data['name'].apply(get_brand_name)
15 # Encoding dictionaries
16 brand_dict = {
17     'Maruti': 1,
18     'Skoda': 2,
19     'Honda': 3,
20     'Hyundai': 4,
21     'Toyota': 5,
22     'Ford': 6,
23     'Renault': 7,
24     'Mahindra': 8,
25     'Tata': 9,
26     'Chevrolet': 10,
27     'Datsun': 11,
28     'Jeep': 12,
29     'Mercedes-Benz': 13,
30     'Mitsubishi': 14,
31     'Audi': 15,
32     'Volkswagen': 16,
33     'BMW': 17,
34     'Nissan': 18,

```

Fig 10.10 Importing libraries & Loading model.pkl & Loading dataset

```

68     'Fourth & Above Owner': 4,
69     'Test Drive Car': 5
70 }
71 #user input
72 name = st.selectbox(
73     'Select Car Brand',
74     cars_data['name'].unique()
75 )
76 year = st.slider(
77     'Car Manufactured Year',
78     1994,
79     2024
80 )
81 km_driven = st.slider(
82     'No of kms Driven',
83     11,
84     200000
85 )
86 fuel = st.selectbox(
87     'Fuel type',
88     cars_data['fuel'].unique()
89 )
90 seller_type = st.selectbox(
91     'Seller type',
92     cars_data['seller_type'].unique()
93 )
94 transmission = st.selectbox(
95     'Transmission type',
96     cars_data['transmission'].unique()
97 )
98 owner = st.selectbox(
99     'Owner type',
100    cars_data['owner'].unique()
101 )

```

Fig 10.11 Creating input fields

```

102 mileage = st.slider(
103     'Car Mileage',
104     10.0,
105     40.0
106 )
107 engine = st.slider(
108     'Engine CC',
109     700.0,
110     5000.0
111 )
112 max_power = st.slider(
113     'Max Power',
114     0.0,
115     200.0
116 )
117 seats = st.slider(
118     'No of Seats',
119     5.0,
120     10.0
121 )
122 # Prediction
123 if st.button("Predict"):
124     # Convert categorical values into numbers
125     name_encoded = brand_dict.get(name)
126     fuel_encoded = fuel_dict.get(fuel)
127     seller_encoded = seller_dict.get(seller_type)
128     transmission_encoded = transmission_dict.get(transmission)
129     owner_encoded = owner_dict.get(owner)
130     # Create input DataFrame
131     input_data_model = pd.DataFrame(
132         [[
133             name_encoded,
134             year,
135             km_driven,

```

Fig 10.12 Creating the input DataFrame & Converting user inputs to numerical values & if st.button("Predict") & model.predict()

```

136     fuel_encoded,
137     seller_encoded,
138     transmission_encoded,
139     owner_encoded,
140     mileage,
141     engine,
142     max_power,
143     seats
144 ]],
145     columns=[
146         'name',
147         'year',
148         'km_driven',
149         'fuel',
150         'seller_type',
151         'transmission',
152         'owner',
153         'mileage',
154         'engine',
155         'max_power',
156         'seats'
157     ]
158 )
159 # Make prediction
160 car_price = model.predict(input_data_model)
161
162 st.success(
163     'Car Price is going to be ₹' + str(round(car_price[0], 2))
164 )
165

```

Fig 10.13 Displaying car price

## 11. Input / Output with Datasets & Supported Screenshots

**Car Price Prediction ML Model**

Select Car Brand  
Tata

Car Manufactured Year  
2018

No of kms Driven  
202235

Fuel type  
Petrol

Seller type  
Trustmark Dealer

Transmission type  
Automatic

Owner type  
First Owner

Car Mileage

Deploy

Fig 11.1 Input

**Car Price Prediction ML Model**

Select Car Brand  
Tata

Car Manufactured Year  
2018

No of kms Driven  
202235

Fuel type  
Petrol

Seller type  
Trustmark Dealer

Transmission type  
Automatic

Owner type  
First Owner

Car Mileage

Predict

Car Price is going to be ₹1253937.17

Fig 11.2 Input & Output

## 12. Images / Video Links

The complete demonstration of the Car Price Prediction using Machine Learning project, including the Streamlit application and prediction process, is available at the following link:

**Video URL:** <https://drive.google.com/file/d/1rOVjiMOCu4emSEVSSunBYH0d1LKkrXE5/view?usp=sharing>

## 13. References

Mayur Dev Sewak  
Head, Internships & Trainings  
Eisystems Services  
&  
Mallika Srivastava  
Head, Training Delivery  
Eisystems Services

## 14. Student Self Evaluation of the Short-Term Internship

Please rate your performance in the following areas:

|                                                     |   |   |   |   |   |
|-----------------------------------------------------|---|---|---|---|---|
| 1) Oral communication                               | 1 | 2 | 3 | 4 | 5 |
| 2) Written communication                            | 1 | 2 | 3 | 4 | 5 |
| 3) Initiative                                       | 1 | 2 | 3 | 4 | 5 |
| 4) Interaction with staff                           | 1 | 2 | 3 | 4 | 5 |
| 5) Attitude                                         | 1 | 2 | 3 | 4 | 5 |
| 6) Dependability                                    | 1 | 2 | 3 | 4 | 5 |
| 7) Ability to learn                                 | 1 | 2 | 3 | 4 | 5 |
| 8) Planning and organization                        | 1 | 2 | 3 | 4 | 5 |
| 9) Professionalism                                  | 1 | 2 | 3 | 4 | 5 |
| 10) Creativity                                      | 1 | 2 | 3 | 4 | 5 |
| 11) Quality of work                                 | 1 | 2 | 3 | 4 | 5 |
| 12) Productivity                                    | 1 | 2 | 3 | 4 | 5 |
| 13) Progress of learning                            | 1 | 2 | 3 | 4 | 5 |
| 14) Adaptability to organization's culture/policies | 1 | 2 | 3 | 4 | 5 |
| 15) OVERALL PERFORMANCE                             | 1 | 2 | 3 | 4 | 5 |

**Rating Scale:** 5 will be Best while 1 will be Worst

**Signature of the Student**

## 15. Annexure 1

### Daily Activity Report

Week No: 1

| Day & Date   | Brief Description of Daily Activity                                                                                                                                                                                                                                                                                    | Learning Outcome                                                                                                                                                                                                                                                                                           | Person In-Charge |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 10 July 2026 | Attended the internship orientation session and learned about the Training, Assessment, and Project (TAP) structure. The session covered the training schedule, attendance rules, Python assessment, project requirements, certification process, email notifications, and professional conduct during the internship. | Understood the internship process, training and attendance requirements, Python assessment, project guidelines, and the conditions required to receive the certificate. Also learned about the importance of checking official emails regularly and maintaining professional behaviour during the program. | Mayur Sewak Dev  |
| 11 July 2026 | It was an extension of an orientation session, and the mentor told us about daily and weekly activity reports.                                                                                                                                                                                                         | We learned about project requirements, certification criteria, official communication through email, and the professional code of conduct expected during the program.                                                                                                                                     | Mayur Sewak Dev  |

## Annexure 1

### Daily Activity Report

Week No: 2

| Day & Date   | Brief Description of Daily Activity                                                                                                                                                                                                                 | Learning Outcome                                                                                                                                                                                                                                                                                                                                                                 | Person In-Charge   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 15 July 2026 | Attended an introductory session on Data Science and Machine Learning, covering the role of data scientists and ML engineers, types of machine learning, labeled and unlabeled data, and the ML lifecycle from data collection to model deployment. | Learned the basics of Data Science and Machine Learning, including structured and unstructured data, labeled and unlabeled data, and the three types of machine learning: supervised, unsupervised, and reinforcement learning. Also understood the basic steps involved in the machine learning process, from collecting and cleaning data to selecting and evaluating a model. | Mallika Srivastava |
| 16 July 2026 | Learned the prerequisites for Machine Learning, set up Jupyter Notebook, and revised Python basics including input, typecasting, data types, string operations, indexing, and Boolean logic.                                                        | Gained basic knowledge of the Python concepts and tools required to start learning Machine Learning.                                                                                                                                                                                                                                                                             | Mallika Srivastava |
| 17 July 2026 | Learned Python lists, including their properties, indexing, nested lists, slicing, mutability, and list methods such as append(), insert(), and extend(). A task was also assigned to document list functions with examples.                        | Gained a clear understanding of how to create, access, slice, modify, and manipulate Python lists, including the difference between commonly used list methods.                                                                                                                                                                                                                  | Mallika Srivastava |
| 19 July 2026 | Studied Python tuples, dictionaries, sets, and list removal methods. Learned their properties, common functions, mutability, copying behavior, and set operations. Also received instructions for submitting the next PDF assignment.               | Gained an understanding of how to use and differentiate Python's major collection data types and their methods, including remove() vs pop(), dictionary access, .copy(), and set operations.                                                                                                                                                                                     | Mallika Srivastava |

## Annexure 1

### Daily Activity Report

Week No: 3

| Day & Date   | Brief Description of Daily Activity                                                                                                                                                                                                                          | Learning Outcome                                                                                                                                              | Person In-Charge   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 20 July 2026 | Learned Python loops, including while and for loops, nested loops, range(), membership operators, and iterating through lists and dictionaries.                                                                                                              | Developed an understanding of loop execution, repetition, pattern printing, sequence iteration, and accessing dictionary keys, values, and items using loops. | Mallika Srivastava |
| 24 July 2026 | Attended a session on Python functions and packages. Learned about default, parameterized, anonymous(lambda) functions, arbitrary arguments, and creating and importing Python packages.                                                                     | Gained practical knowledge of Python functions, packages, and module importing for code reusability.                                                          | Mallika Srivastava |
| 25 July 2026 | Attended a session on the NumPy library. Learned about NumPy arrays, different array dimensions (1D, 2D, 3D), array properties, and various methods to create arrays such as array(), arrange(), linspace(), logspace(), zeros(), ones(), and random arrays. | Gained practical understanding of NumPy array creation, array properties, and commonly used methods for data handling in Python.                              | Mallika Srivastava |
| 26 July 2026 | Our instructor reviewed our reports and provided corrections and guidance on the correct way to prepare them.                                                                                                                                                | Learned how to prepare the report correctly and understood the changes needed to improve its format and presentation.                                         | Mallika Srivastava |

## Annexure 1

### Daily Activity Report

Week No: 4

| Day & Date   | Brief Description of Daily Activity                                                                                                                                                                                                              | Learning Outcome                                                                                                                                                 | Person In-Charge   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 27 July 2026 | Learned about NumPy 2D arrays, including how access rows, columns, and individual elements. Also studied array properties such as ndim, shape, and size, and learned how to reshape arrays.                                                      | Gained an understanding of accessing, and reshaping 2D NumPy arrays and using array properties to check their dimensions, shape, and size.                       | Mallika Srivastava |
| 30 July 2026 | Learned NumPy 3D arrays, including their shape, dimensions, and slicing. Practiced accessing rows, columns, and elements, reshaping arrays, and using functions such as arange(), reshape(), vsplit(), hsplit(), and concatenate().              | Gained an understanding of creating, accessing, slicing, reshaping, splitting, and combining NumPy arrays.                                                       | Mallika Srivastava |
| 01 Aug 2026  | Learned the basics of Pandas, including creating a Pandas Series and DataFrame, accessing rows and columns, using indexing, and checking data information. Also studied functions such as head(), tail(), count(), max(), min(), and describe(). | Gained an understanding of working with Pandas Series and DataFrames, accessing and checking data, and using basic functions to view and understand the dataset. | Mallika Srivastava |

## Annexure 1

### Daily Activity Report

Week No: 5

| Day & Date  | Brief Description of Daily Activity                                                                                                                                                                                                                                                                                                                                                                                                                                          | Learning Outcome                                                                                                                                             | Person In-Charge   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 03 Aug 2026 | Reviewed Pandas assignments and learned how to load CSV files into Jupyter/Colab. Practiced DataFrame operations such as viewing data, selecting rows and columns, using .loc and .iloc, applying filters with multiple conditions, and using between() and isin(). A Netflix dataset was also provided for practice.                                                                                                                                                        | Gained an understanding of importing CSV files, accessing and filtering DataFrame data using different Pandas methods, and exploring datasets independently. | Mallika Srivastava |
| 06 Aug 2026 | The class covered data grouping using groupby() and get_group(), random data sampling with sample(), frac, and random_state, DataFrame merging using concat() and pd.merge() with different join types, and basic data cleaning by replacing invalid values with np.nan. The instructor also explained the difference between temporary and permanent changes using inplace=True and introduced missing value handling, which will be covered in detail in the next session. | Gained an understanding of grouping, sampling, merging DataFrames, and replacing invalid values as part of the data cleaning process.                        | Mallika Srivastava |
| 07 Aug 2026 | Learned how to handle missing values in Pandas using isnull(), dropna(), and fillna(). Also learned categorical data encoding and practiced data cleaning using the Titanic dataset.                                                                                                                                                                                                                                                                                         | Gained an understanding of detecting, removing, and filling missing data, along with converting categorical data into numeric values for Machine Learning.   | Mallika Srivastava |
| 09 Aug 2026 | Learned Matplotlib for data visualization, including creating and customizing line, scatter, bar, and pie charts. Practiced adding labels, titles, legends, grids, and saving plots.                                                                                                                                                                                                                                                                                         | Gained an understanding of creating and customizing different types of plots to visualize and compare data effectively.                                      | Mallika Srivastava |

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## Annexure 1

### Daily Activity Report

Week No: 6

| Day & Date  | Brief Description of Daily Activity                                                                                                                                                         | Learning Outcome                                                                                                                                                  | Person In-Charge   |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 13 Aug 2026 | Learned about Matplotlib subplots and practiced creating multiple graphs in one figure. Worked with Pandas and the Netflix dataset to clean data, find counts, and create different charts. | Gained an understanding of creating subplots, using <code>value_counts()</code> , cleaning data, and visualizing Netflix data using bar, scatter, and pie charts. | Mallika Srivastava |

Week No: 7

| Day & Date  | Brief Description of Daily Activity                                                                                                                                                                                      | Learning Outcome                                                                                                                                          | Person In-Charge   |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 18 Aug 2026 | Reviewed Matplotlib tasks and data cleaning, then started Machine Learning modeling with simple and multiple linear regression using practical datasets.                                                                 | Learned the basics of linear regression, including slope and intercept, and gained an understanding of using Scikit-learn for building regression models. | Mallika Srivastava |
| 20 Aug 2026 | Learned about classification using an insurance dataset and studied Logistic Regression to predict whether a person will buy insurance based on age. Practiced train-test splitting and model evaluation using accuracy. | Understood the basics of classification, Logistic Regression, sigmoid function, training and testing data, predictions, and accuracy.                     | Mallika Srivastava |

## Annexure 1

### Daily Activity Report

Week No: 8

| Day & Date  | Brief Description of Daily Activity                                                                                                                                                           | Learning Outcome                                                                                                                                   | Person In-Charge   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 21 Aug 2026 | Learned about handwritten digit recognition using the 8×8 digits dataset. Practiced reshaping image data, preparing features, and building a Logistic Regression model using train-test data. | Understood how image data is converted into numerical features and how Logistic Regression can be used for digit classification and prediction.    | Mallika Srivastava |
| 22 Aug 2026 | Learned about image dataset preparation using OpenCV, Pillow, NumPy, and CIFAR-10. Studied the Iris dataset and practiced K-Nearest Neighbors for multi-class classification.                 | Gained an understanding of image preprocessing, data preparation, model training, and evaluation using K-Nearest Neighbors.                        | Mallika Srivastava |
| 25 Aug 2026 | Worked on an image classification project using OpenCV and NumPy. Learned SVM, K-fold cross-validation, and hyperparameter tuning using Grid Search CV.                                       | Gained an understanding of image preprocessing, SVM classification, cross-validation, and selecting suitable model parameters to improve accuracy. | Mallika Srivastava |

Week No: 9

| Day & Date  | Brief Description of Daily Activity                                                                                                                                                                                    | Learning Outcome                                                                                                                            | Person In-Charge   |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 29 Aug 2026 | Learned about model deployment and how to save a trained model as a pickle file for future use. The instructor also explained the project report and submission process, cleared doubts, and concluded the internship. | Understood the basic process of deploying a trained model, preparing the project report, and following the project submission instructions. | Mallika Srivastava |

## 16. Annexure 2

### Weekly Progress Report

| Week(s) | Summary of Weekly Activity                                                                                                                                                                                                                                                                                                                                                                                                |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1  | During the week, I attended internship orientation sessions and learned about the TAP structure, training schedule, attendance, Python assessment, project requirements, certification criteria, daily and weekly activity reports, official email communication, and professional conduct expected during the internship.                                                                                                |
| Week 2  | During the week, I was introduced to the basics of Data Science and Machine Learning, including different types of ML and the ML process. I also set up Jupyter Notebook and revised Python basics. Further, I learned about lists, tuples, dictionaries, and sets, along with their properties, methods, indexing, slicing, mutability, and common operations.                                                           |
| Week 3  | During the week, I learned about Python loops, functions, packages, and the NumPy library. I practiced different types of loops, functions, package importing, and NumPy array creation and properties. The instructor also reviewed our reports and provided corrections and guidance to improve their format and presentation.                                                                                          |
| Week 4  | During the week, I learned about NumPy 2D and 3D arrays, including accessing elements, rows, and columns, slicing, reshaping, splitting, and combining arrays. I also learned the basics of Pandas, including creating Series and DataFrames, accessing rows and columns, checking data information, and using functions such as head(), tail(), count(), max(), min(), and describe().                                   |
| Week 5  | During the week, I learned Pandas data handling and cleaning, including loading CSV files, selecting and filtering data, grouping, sampling, merging DataFrames, and handling missing values. I also practiced categorical data encoding using datasets such as Netflix and Titanic. Additionally, I learned Matplotlib and practiced creating and customizing line, scatter, bar, and pie charts for data visualization. |

## Annexure 2

### Weekly Progress Report

| Week(s) | Summary of Weekly Activity                                                                                                                                                                                                                                                                                                                                          |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 6  | During the week, I learned about Matplotlib subplots and practiced creating multiple graphs in one figure. I also worked with Pandas and the Netflix dataset to clean data, find counts using <code>value_counts()</code> , and create bar, scatter, and pie charts for data visualization.                                                                         |
| Week 7  | During the week, I reviewed Matplotlib and data cleaning and started learning Machine Learning modeling. I practiced simple and multiple linear regression using practical datasets and learned the basics of Logistic Regression for classification. I also learned about train-test splitting, predictions, sigmoid function, and model accuracy.                 |
| Week 8  | During the week, I learned about image classification and Machine Learning models. I practiced digit classification using Logistic Regression and learned image data preparation with OpenCV, Pillow, and NumPy. I also studied K-Nearest Neighbors (KNN), SVM, K-fold cross-validation, and Grid Search CV for model training, evaluation, and improving accuracy. |
| Week 9  | During the week, I learned about model deployment and saving trained models as pickle files for future use. I also learned about the project report and submission process, received guidance on my doubts, and completed the internship training.                                                                                                                  |